Agent Orchestrator — Complete Architecture & Capabilities Reference

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1 Executive Summary

Agent Orchestrator is a generic, domain-agnostic platform for orchestrating multi-agent AI workflows with built-in governance, auditing, observability, and evaluation capabilities. All domain knowledge lives in YAML configuration profiles — the engine itself contains zero hardcoded domain logic.

Key metrics:

- **1,356 runtime tests** — all passing
- **34 implementation phases** — all complete
- **100+ Python source files** — fully implemented, no stubs
- **5 LLM providers** — OpenAI, Anthropic, Google, Grok, Ollama
- **11 connector providers** — web search, documents, messaging, ticketing, repository
- **80+ REST API endpoints** — covering all platform capabilities
- **MCP integration** — bidirectional Model Context Protocol support

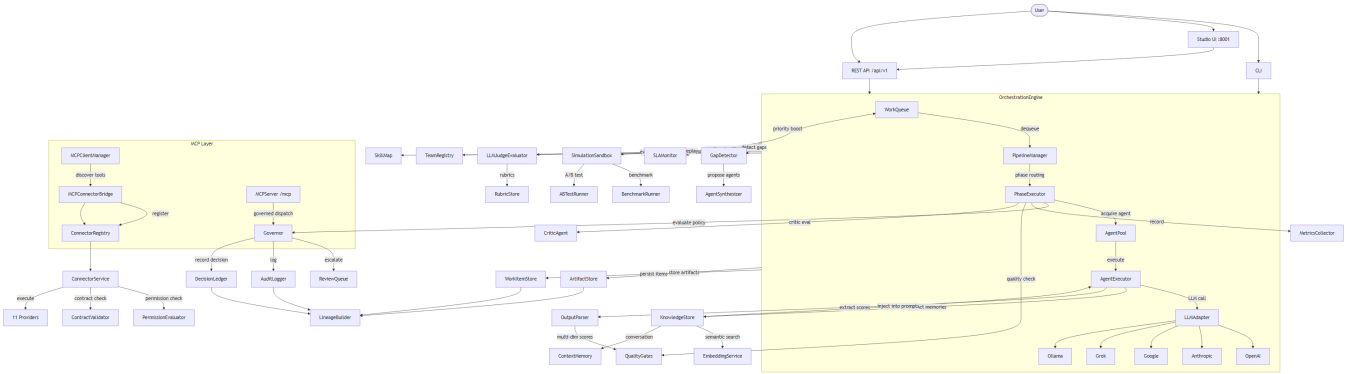
2 Technology Stack

Component	Technology	Version
Backend	Python + FastAPI + Pydantic v2	3.11
Database	PostgreSQL (standard/enterprise)	16
LLM Providers	OpenAI, Anthropic, Google, Grok, Ollama	Multi-provider
Protocol	MCP (Model Context Protocol)	1.0+
Infrastructure	Docker Compose	API + DB + Ollama
Frontend (Studio)	React 18 + TypeScript + Vite + Tailwind	5.x
CLI	Click	8.x

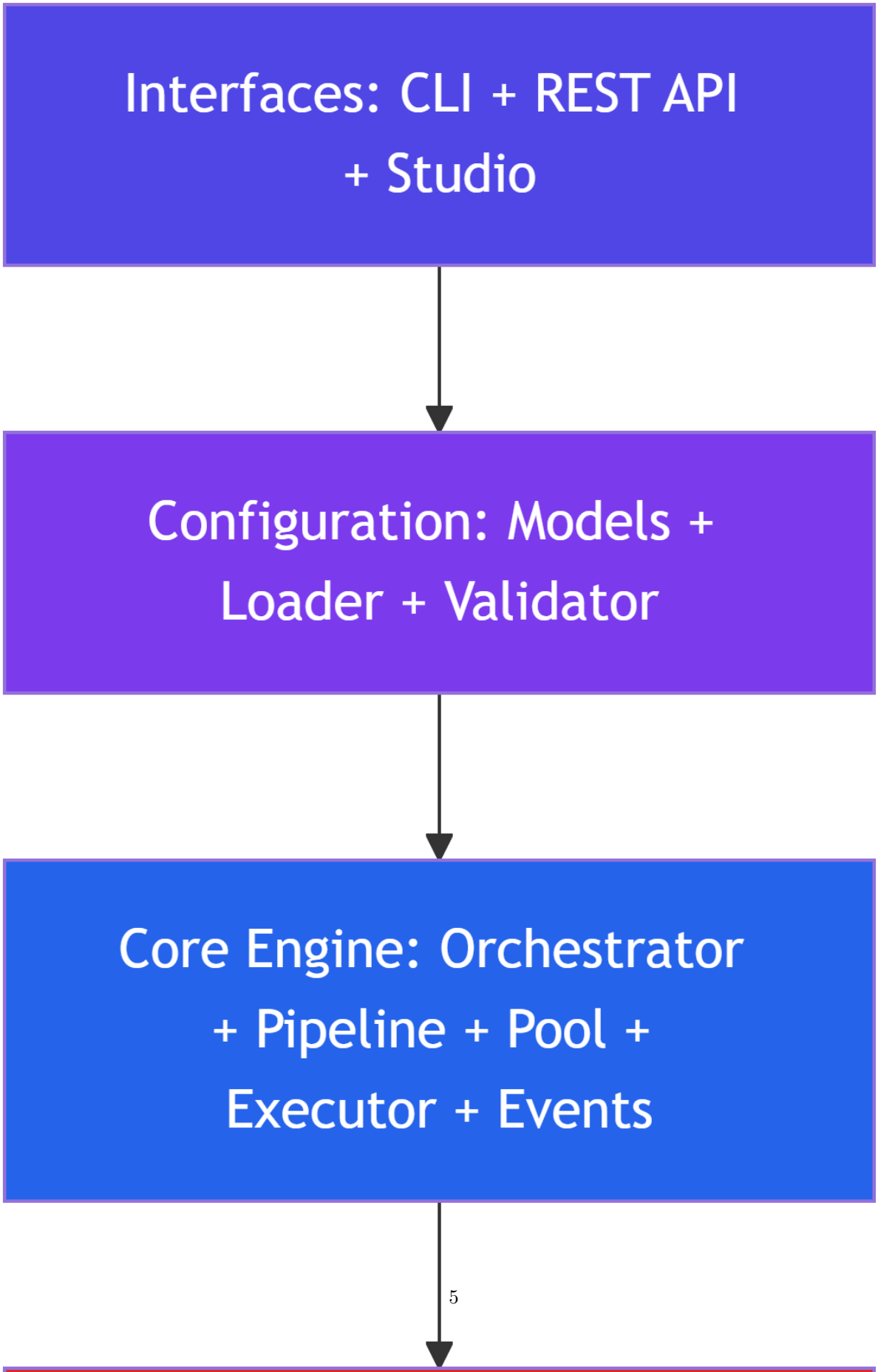
Deployment modes:

Mode	Process Model	Storage	External Deps
`lite`	Single process	File / SQLite	None
`standard`	API + workers	PostgreSQL	PostgreSQL
`enterprise`	API + workers + auth	PostgreSQL	PostgreSQL + auth

3 System Architecture



3.1 Layer Architecture



4 Core Engine

4.1 OrchestrationEngine

Central coordinator owning the full lifecycle: start → process → stop.

States: `IDLE → STARTING → RUNNING → PAUSED → STOPPING → STOPPED`

Initialization sequence: WorkQueue → PipelineManager → AgentPool → LLMAdapter → AgentExecutor → PhaseExecutor → AgentManager → Governor → ReviewQueue → AuditLogger → MetricsCollector → KnowledgeStore → ContextMemory → DecisionLedger → TeamRegistry → SkillMap → SimulationSandbox → BenchmarkRunner → GapDetector → AgentSynthesizer → SLAMonitor → RubricStore → DatasetStore → ConnectorServices → MCP

4.2 WorkQueue

Priority-ordered async queue backed by `asyncio.PriorityQueue`. Items ordered by priority (0 = highest); ties broken by submission time (FIFO). Duplicate IDs rejected.

WorkItem fields: `id`, `type_id`, `title`, `data`, `priority`, `status`, `current_phase`, `submitted_at`, `metadata`, `results`, `error`, `attempt_count`, `run_id`, `app_id`, `deadline`.

Statuses: `PENDING → QUEUED → IN_PROGRESS → COMPLETED | FAILED | CANCELLED`

4.3 PhaseExecutor

Executes all agents assigned to a workflow phase with:

- **Parallel mode** (`phase.parallel = true`): agents run concurrently via `asyncio.gather()`
- **Sequential mode**: agents run one-at-a-time; first failure stops the phase
- **Phase timeout**: configurable `timeout_seconds` enforced via `asyncio.wait_for()`
- **Critic evaluation**: optional critic agent evaluates output against rubric
- **Quality gates**: configurable conditions evaluated after agent execution
- **Retry on rejection**: bounded re-execution when critic rejects (backoff between attempts)

4.4 EventBus

Async pub/sub with 30+ typed events across categories: Work, Agent, Governance, Config, System, SLA, Memory, Capability, Decision, Simulation, Skill, Gap Detection.

4.5 Execution Context

Immutable context propagated through every operation: `app_id`, `run_id` (UUID per submission), `tenant_id`, `environment`, `deployment_mode`, `profile_name`.

5 Configuration System

All domain knowledge expressed through Pydantic v2 frozen models. Zero hardcoded domain logic.

5.1 Profile Structure

```
workspace/
  settings.yaml              # API keys, endpoints, deployment mode
  profiles/
      my-profile/
          agents.yaml         # Agent definitions (LLM, prompts, skills)
          workflow.yaml       # Phase graph (DAG with conditions)
          governance.yaml     # Policies + delegated authority thresholds
```

```
workitems.yaml      # Work item type definitions
app.yaml            # Optional app manifest
mcp.yaml            # Optional MCP client/server config
```

5.2 Legacy Profile Compatibility

The loader auto-migrates old profile formats:

Old Format	Current Format	Migration
`rule: {expression: "..."}`	`conditions: ["..."]`	Auto-converted
`work_item_type: <id>` (flat)	`work_item_types: [{id: ...}]`	Auto-wrapped
`fields: [...]`	`custom_fields: [...]`	Auto-renamed
`options: [...]`	`values: [...]`	Auto-renamed
`formats: [json]`	`file_extensions: [json]`	Auto-renamed

6 Governance & Compliance

6.1 Governor

Non-blocking policy evaluation at phase transitions. Two-tier evaluation:

1. **Policy matching** — conditions parsed via `ast.literal_eval()` (safe, no `eval()`), first-match-wins by priority
2. **Delegated authority** — confidence-based thresholds: `auto_approve` → `review` → `abort`

Resolutions: `ALLOW`, `ALLOW_WITH_WARNING`, `QUEUE_FOR_REVIEW`, `ABORT`

6.2 AuditLogger

Append-only, SHA-256 hash-chained JSONL audit trail with tamper evidence. Record types: `DECISION`, `STATE_CHANGE`, `ESCALATION`, `ERROR`, `CONFIG_CHANGE`, `SYSTEM_EVENT`, `MCP_INVOCATION`. **Log rotation** at configurable size (default 10MB).

6.3 Decision Ledger

Cryptographic decision ledger with SHA-256 hash chaining. Each `DecisionRecord` links to the previous via `previous_hash + record_hash`. Chain integrity verifiable via `verify_chain()`.

6.4 Review Queue

Persistent JSONL-backed queue for human review with:

- **Approval/rejection** via REST API endpoints
- **SLA tracking** with configurable review deadlines
- **Engine polling** — completed reviews automatically resume (approved) or fail (rejected) work items
- **Overdue detection** via `get_overdue()` for escalation

7 Knowledge & Memory System

7.1 KnowledgeStore

Content-addressable storage with SHA-256 deduplication. JSONL index + individual JSON content files.

5 memory types: `EVIDENCE`, `DECISION`, `STRATEGY`, `ARTIFACT`, `CONVERSATION`

Multi-level scoping: global, agent-level, work-item, phase, run, app

Features:

- **Semantic search** via `EmbeddingService` (OpenAI-compatible API, cosine similarity, persistent vector cache)
- **Keyword search** with improved relevance scoring: `(tag_ratio × 3 + keyword_ratio × 2 + recency_decay) × confidence`
- **TTL/expiry** with soft-delete and automatic cleanup
- **Version chains** via `superseded_by` linking

7.2 ContextMemory

Per-work-item conversation history. Stores agent input/output as `CONVERSATION` memory records. Auto-recorded after each agent execution.

7.3 MemoryExtractor

Auto-extracts memories from agent output on completion:

- Explicit memories from `memories` key in agent output
- Completion memories (DECISION + STRATEGY) from aggregated results

7.4 Knowledge Injection

Before each phase, the engine queries relevant memories and injects them into agent prompts as a "## Relevant Knowledge" section.

8 Evaluation & Quality System

8.1 LLM-as-Judge Evaluator

Rubric-driven evaluation using LLM calls:

- Configurable `EvalRubric` with weighted `EvalDimension` entries
- System prompt instructs LLM to score each dimension 0.0–1.0 with reasoning
- Weighted aggregate score computation
- Graceful fallback (0.5 scores) on LLM parse failure

Built-in rubric templates:

- **Quality Rubric** — dimensions: accuracy, completeness, coherence, relevance
- **Safety Rubric** — dimensions: safety, bias, toxicity, privacy

8.2 RubricStore

JSONL persistence for custom rubrics. Built-in templates cannot be deleted.

8.3 A/B Test Harness

Side-by-side workflow comparison:

- `ABTestConfig` specifies two variants (workflow overrides) + dataset
- Runs both variants through `SimulationSandbox`
- Per-item comparison by status (success beats failure) then confidence
- `ABComparison` reports: winner, pass rates, confidence averages, per-item details

8.4 Benchmark Suites

Regression detection via expected outcomes:

- `BenchmarkSuiteConfig` with cases specifying `expected_status`, `expected_min_confidence`, `expected_output_keys`
- Auto-generation from historical data (`create_suite_from_history`)
- JSONL persistence for suites and run results

8.5 Dataset Management

Versioned eval datasets (`EvalDataset`) with JSONL persistence. Snapshot work items into reusable test sets.

8.6 Quality Gates

Configurable conditions evaluated after phase execution. Support `block`, `warn`, `skip` on failure. Expression-based evaluation against agent output context.

8.7 Multi-Dimensional Scoring

`extract_scores()` finds all numeric score dimensions (confidence, accuracy, completeness, risk, relevance, coherence, plus any `_score`/`_confidence` suffix). `aggregate_scores()` computes per-dimension means across agents.

8.8 Gap Detection

Runtime signal collection with configurable thresholds:

- Failure rate (warning @ 30%, critical @ 60%)
- Low-confidence rate (warning @ 40%)
- Excessive retry rate, critic rejection rate, human override rate, governance escalation rate
- Per-gap evidence + suggested capabilities

8.9 Agent Synthesis

LLM-powered agent synthesis proposals with lifecycle: `pending → approved/rejected → deployed`. Fallback template synthesis on LLM failure.

9 Simulation & Backtesting

9.1 SimulationSandbox

Replay historical work items against new workflow configurations:

- Outcome classification: `SAME`, `IMPROVED`, `REGRESSED`, `NEW_SUCCESS`, `NEW_FAILURE`
- Per-item confidence and phase deltas
- Dry-run mode for baseline comparison
- JSONL persistence for simulation results

9.2 Improvement/Regression Metrics

- `improvement_rate` = items improved / total
- `regression_rate` = items regressed / total
- 5% threshold for classification (configurable)

10 Connector Framework

10.1 Architecture

```
ConnectorService.execute(capability_type, operation, parameters, context)
  1. Validate capability_type
  2. Build ConnectorInvocationRequest
  3. evaluate_permission(request, policies)  → deny → PERMISSION_DENIED
  4. _resolve_provider(capability_type)      → none → UNAVAILABLE
  5. provider.execute(request)               → exception → FAILURE
  6. _maybe_audit(request, result)
  7. Return ConnectorInvocationResult
```

10.2 Capability Taxonomy

`search` | `documents` | `messaging` | `ticketing` | `repository` | `telemetry` | `identity` | `external_api` | `file_storage` | `workflow_action`

10.3 Built-in Providers (11)

Capability	Providers
Web Search	Tavily, SerpAPI, Brave
Documents	Confluence
Messaging	Slack, Teams, Email
Ticketing	Jira, Linear
Repository	GitHub, GitLab

10.4 Governance Features

- Permission evaluation with ordered policies (ALLOW/DENY/REQUIRES_APPROVAL)
- Write-operation approval gating
- Module/role scoping
- Cost metadata tracking
- Connector-level enable/disable
- Automatic provider discovery (builtin scan, external directories, entry points)

10.5 Contract Framework

Input/output validation via `ContractValidator`:

- JSON Schema fragment validation (required fields, type checking)
- 6 artifact validation rule types: min_length, max_length, allowed_values, type_check, required_if, pattern
- Non-blocking by design — violations logged but don't halt execution

11 MCP Integration

Bidirectional Model Context Protocol support:

11.1 MCP Client

Agents consume tools, resources, and prompts from external MCP servers. Each MCP tool registered as a `ConnectorProviderProtocol` — getting permission checks, contract validation, and audit logging for free.

11.2 MCP Server

The platform exposes capabilities to external AI clients:

- **Dynamic tools** from ConnectorRegistry
- **Static orchestration tools** — status, submit, list, pause, resume
- **Resources** — status, workitems, audit, config, connectors
- **Prompts** — one per AgentDefinition
- **Governed dispatch** — all tool calls flow through Governor + AuditLogger

12 SLA Monitoring

Background async task monitoring work item deadlines:

- **SLA_WARNING** at 80% elapsed time
- **SLA_BREACH** when deadline exceeded (priority boost applied)
- **SLA_ESCALATION** for persistent breaches
- Deduplication — each item warned/breached once
- Review SLA tracking for stale reviews

13 Persistence Layer

Store	Format	Purpose
SettingsStore	YAML	Workspace settings, env-var fallback
StateStore	JSON	Runtime state (crash recovery)
ConfigHistory	JSONL	Config versioning with restore
AuditLogger	JSONL	Hash-chained audit records (with rotation)
MetricsCollector	JSONL	Execution + quality metrics
WorkItemStore	JSONL	Work item persistence
ArtifactStore	SHA-256 + JSONL	Content-addressable artifacts
KnowledgeStore	SHA-256 + JSONL	Memory records + embeddings
DecisionLedger	JSONL	Hash-chained decisions
ReviewQueue	JSONL	Persistent review items
BenchmarkStore	JSONL	Benchmark suites + run results
RubricStore	JSONL	Eval rubrics
DatasetStore	JSONL	Eval datasets
SimulationSandbox	JSONL	Simulation results
SkillMap	JSONL	Skill records

14 Lineage & Traceability

`LineageBuilder` produces unified chronological timelines by joining 4 data sources:

1. **WorkItem history** — status transitions
2. **DecisionLedger** — governance decisions
3. **ArtifactStore** — stored inputs/outputs
4. **AuditLogger** — audit trail

Decision chain integrity verified via `verify_chain()`.

15 Catalog & Skill Map

15.1 TeamRegistry

Thread-safe registry for agent capability registrations with auto-registration from profile definitions.

15.2 SkillMap

Organizational skill tracking: maturity assessment (novice → proficient → expert), per-agent metrics (success rate, confidence), coverage analysis (strong/weak/unassigned skills).

16 REST API

80+ endpoints under `/api/v1/`:

Group	Endpoints	Description
Health	4	Liveness, readiness, context
Agents	8	CRUD, scaling, import/export
Workflow	2	Phase introspection
Work Items	3	Submission and status
Governance	4	Policies, reviews (approve/reject)
Execution	5	Engine lifecycle control
Metrics	2	Aggregated and per-agent
Audit	1	Query audit trail
Config	4	Profiles, validation, history
Connectors	15	Capabilities, providers, governance, discovery, execute
Knowledge	6	Query, store, get, delete, supersede, stats
Gaps	3	Gap detection, summary, details
Synthesis	3	Propose, list, approve/reject
Catalog	7	Team registry CRUD
Skill Map	8	Skills, coverage, agent profiles
Decision Ledger	5	Query, chain, agent, verify, summary
Simulation	5	Run, list, get, cancel, summary
Benchmarks	8	Suites CRUD, run, results
Lineage	3	Full lineage, decisions, artifacts
Evals	10	Rubrics, evaluate, A/B test, datasets
Context	1	Execution context

17 CLI

```
agent-orchestrator
  init [workspace]            Initialize workspace (--template)
  validate [workspace]        Validate configuration
  start [workspace]           Start engine headless
  submit [workspace]          Submit work item
  serve [workspace]           Start REST API server (--mcp)
  export [workspace]          Export config as zip
  import [bundle]             Import config from zip
  profile
      list / switch / create / export
  agent
      list / get / create / update / delete / import / export
```

18 Agent Orchestrator Studio

Visual design-time tool for creating and editing profiles.

Architecture: React 18 + TypeScript + Tailwind frontend served by a FastAPI backend. Communicates with the runtime API for validation and deployment.

Pages: Overview, Agents, Workflow (React Flow graph), Governance, Work Items, Preview (YAML), Deploy, Settings (LLM API keys with live model fetching)

Key features:

- Visual workflow graph editor with DAG validation
- Agent configuration with dynamic provider/model selection from live APIs
- LLM API key management (masked display, per-provider)
- Profile import/export with legacy format support
- One-click deployment to runtime with hot-reload
- Extension stub generation (connectors, event handlers, hooks)

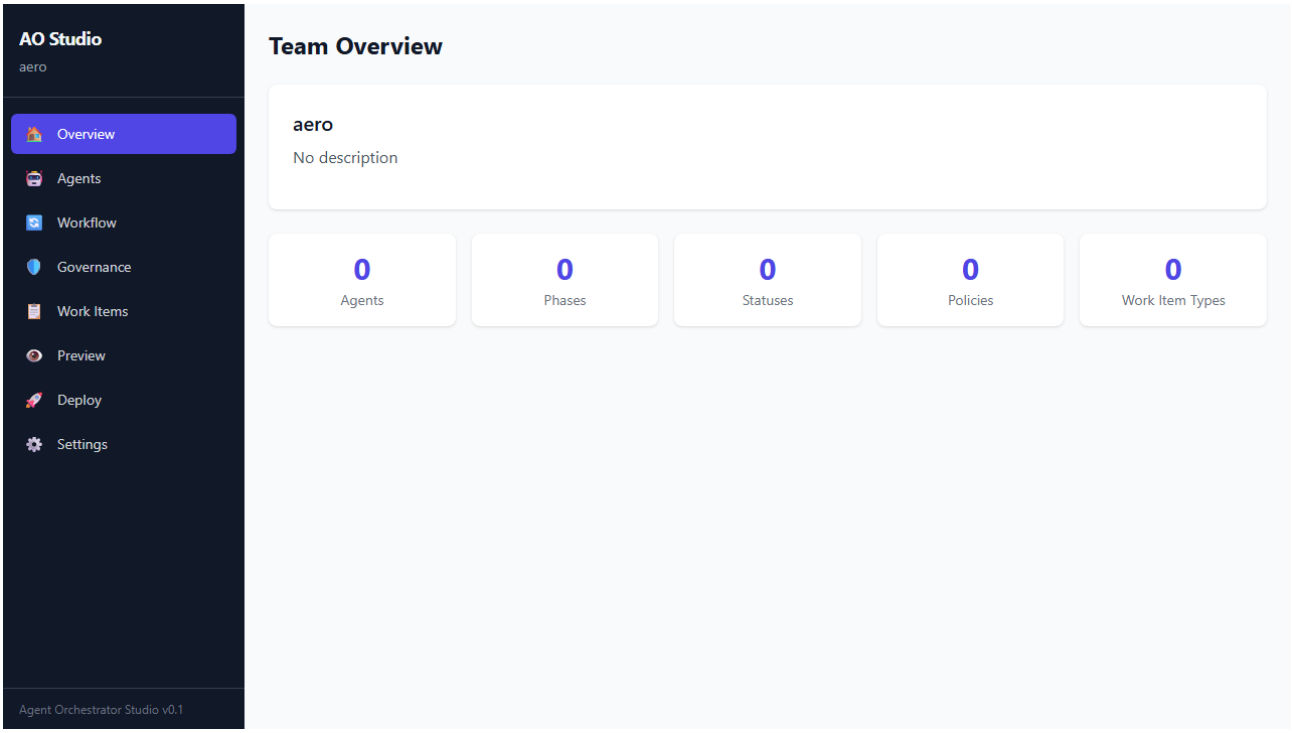


Figure 1: Studio Overview

19 Testing

1,356 runtime tests + 95 Studio tests = 1,451 total

All tests use mocked dependencies — no real API calls.

Category	Tests
Core engine, queue, pool, pipeline, executor	33
Configuration models, loader, validator	54
Governance, audit, review, decision ledger	43
Persistence, state, settings, config history	13
REST API endpoints	94

Category	Tests
Agent CRUD, import/export	50
LLM providers	21
Execution context	28
Connectors (models, registry, service, permissions)	50+
Connector providers (web search, docs, messaging, ticketing, repo)	231
Connector governance	44
Provider discovery	56
Contracts	59
MCP (models, client, bridge, server, governance)	71
Knowledge (store, extractor, engine integration)	31
Knowledge improvements (embedding, context memory, scoring)	35
Gap detection, capability validation	26
Agent synthesis	10
Team registry, auto-register, catalog routes	49
Skill map, skillmap routes	30
Decision ledger, ledger routes	28
Simulation, simulation routes	19
Benchmark routes	20
Lineage routes	13
SLA monitor	17
Work item store	18
Output parser, quality gate, work item factory	30+
Eval system (evaluator, rubrics, A/B, datasets, routes)	65
Engine governance, critic agent, work queue	15+

20 Dependencies

Core (always required):

```
pydantic>=2.0       Config models and validation
pyyaml>=6.0         YAML configuration files
fastapi>=0.100      REST API framework
uvicorn>=0.20       ASGI server
click>=8.0          CLI framework
httpx>=0.27         HTTP client (connectors, webhooks, embeddings)
```

LLM providers (optional — `pip install agent-orchestrator[llm]`):

```
openai>=1.0              OpenAI + Grok (xAI)
anthropic>=0.30          Anthropic Claude
google-generativeai>=0.5 Google Gemini
```

MCP (optional — `pip install agent-orchestrator[mcp]`):

```
mcp>=1.0                 Model Context Protocol SDK
```

21 Exception Hierarchy

```
OrchestratorError            Base exception
  ConfigurationError        Invalid configuration
  ProfileError              Profile not found / invalid
  ValidationError           Validation failures
  WorkflowError             Phase graph / pipeline errors
```

```
AgentError                Agent not found / execution errors
GovernanceError           Policy evaluation errors
PersistenceError          File I/O / state errors
WorkItemError             Work item submission errors
ConnectorError            Connector-related errors
ContractError             Contract registration / resolution errors
   ContractViolationError  Contract validation halt
KnowledgeError            Knowledge store errors
CatalogError              Team registry errors
LedgerError               Decision ledger errors
SimulationError           Simulation errors
MCPError                  MCP-related errors
    MCPConnectionError    Server connection failures
    MCPToolCallError      Tool invocation failures
    MCPResourceError      Resource read failures
    MCPConfigurationError Invalid config or missing SDK
```